

Project

- [GITHUB Repo](#)

Competition Rules

1. Talk about the functionality and importance of these parts explicitly citing each corresponding rule from the competition [rulebook](#) :

- Brake System Plausibility Device
- Insulation Monitoring Device
- Accumulator Isolation Relay
- Brake over travel switch
- Inertia switch
- Tractive system active light

1. Brake System Plausibility Device (BSPD)

- **Functionality:** The BSPD is a standalone, non-programmable circuit directly supplied by the LVMS ([T11.6.3](#), [T11.6.4](#)) . It continuously monitors braking pressure via a brake system pressure sensor ([T11.6.5](#)) and motor power via a DC current sensor ([T11.6.6](#)). It automatically opens the shutdown circuit when hard braking occurs while 5 kW or more is being delivered to the motors ([T11.6.1](#)) and this condition persists for more than 500 ms ([T11.6.2](#)).
- **Importance:** Prevents unintended acceleration and runaway motor power during hard braking scenarios, providing an independent, fail-safe safety mechanism without relying on software control.

2. Insulation Monitoring Device (IMD)

- **Functionality:** According to [EV6.3.3](#), every vehicle must have an IMD installed in the Tractive System (TS) to continuously monitor insulation resistance between the high-voltage system and the chassis, set to a response value of $\geq 500 \Omega/V$ of maximum TS voltage. The device must be connected on the vehicle side of the AIRs ([EV6.3.4](#)) and maintain separate chassis-ground connections to the grounded TSAC and main hoop ([EV6.3.5](#)). It opens the shutdown circuit during an insulation fault, open measurement connection, or IMD failure without

using programmable logic ([EV6.3.6](#)), and triggers a clearly visible red cockpit indicator marked "IMD" ([EV6.3.7](#)).

- **Importance:** Protects the driver, pit crew, and track marshals from electrical shock hazards by ensuring that any insulation failure or loss of ground integrity immediately de-energizes the vehicle.

3. Accumulator Isolation Relays (AIRs)

- **Functionality:** According to [EV5.6.1](#), at least two AIRs must be installed inside each Tractive System Accumulator Container (TSAC). When opened, they disconnect both poles of the TS accumulator so that no TS voltage is present outside the container and the vehicle side is galvanically isolated ([EV5.6.2](#)). Rule [EV5.6.3](#) mandates that the AIRs must be mechanical, normally-open relays (solid-state relays are strictly prohibited).
- **Importance:** Acts as the primary physical barrier preventing high-voltage energy from exiting the accumulator container when the vehicle is deactivated or when a safety shutdown occurs.

4. Brake Over-Travel Switch (BOTS)

- **Functionality:** A mechanical switch incorporated into the shutdown circuit and mounted near the brake pedal assembly. If mechanical or hydraulic brake failure causes the brake pedal to travel beyond its intended limit, the pedal mechanically actuates the switch to immediately open the shutdown circuit without relying on software control.
- **Importance:** Provides an essential mechanical fail-safe that shuts down motor power when mechanical or hydraulic brake failure compromises vehicle stopping capability.

5. Inertia Switch

- **Functionality:** According to [T11.5.1](#), an inertia switch must be integrated into the shutdown circuit to automatically open and latch until manually reset upon an impact. Rule [T11.5.2](#) requires it to trigger at an omnidirectional peak acceleration of ≤ 8 g for a half-sine pulse of ≥ 50 ms or ≤ 13 g for ≥ 20 ms. The switch must contain no semiconductor components ([T11.5.3](#)) and must be rigidly attached to the vehicle according to manufacturer specifications while remaining removable for testing ([T11.5.4](#)).
- **Importance:** Provides an independent mechanical crash-detection mechanism that instantly isolates high-voltage systems after a collision, mitigating fire and electrical hazard risks for responders.

6. Tractive System Active Light (TSAL)

- **Functionality:** Rule [EV4.10.1](#) requires a single TSAL that clearly indicates TS status and performs no other function. Per [EV4.10.2](#), it must display a flashing red light (2–5 Hz) whenever dangerous TS voltage is present or active (remaining active for the required duration after LVS is turned off). A continuous green light indicates that the Tractive System is safe, with AIRs and pre-charge relays open and voltage below safe thresholds ([EV4.10.3](#)). Per [EV4.10.9](#), the TSAL must operate using hard-wired electronics without software control, positioned according to visibility requirements ([EV4.10.7](#), [EV4.10.8](#)).
- **Importance:** Provides immediate visual safety communication to anyone approaching or working on the vehicle, preventing accidental contact with an energized Tractive System.

PCB & Circuits

2. What is Crosstalk in PCB and How to reduce it?

-Crosstalk is unwanted interference that occurs when a signal from one PCB trace affects a nearby trace due to electromagnetic coupling. It is mainly caused by capacitive and inductive coupling between closely spaced traces, especially when signals change rapidly.

Ways to reduce crosstalk:

- ☐ Increase the spacing between PCB traces.
- ☐ Avoid routing traces parallel to each other for long distances.
- ☐ Use a continuous ground plane as a reference layer.
- ☐ Use proper PCB routing and impedance control.
- ☐ Reduce signal rise and fall times when possible.

3. Build this circuit in your simulator and answer the following questions:

[LRC Circuit - Circuit Simulator](#)

a) What is this circuit's primary function?

- The primary function of this circuit is to show how 50Hz electrical noise from wall outlets interferes with electronics, and how using a low resistance or a closed switch can block that noise.

b) What do the components inside the dashed box represent? Explain in detail.

- The dashed box represents ambient electrical interference (stray noise) coupling from nearby power lines into the circuit through the air.

- 50Hz Source: This represents the AC voltage from standard wall outlets and power cables in the room.
- 1GΩ Resistor (R3): This represents the extremely high resistance of the air gap between the power line and the circuit. In real life, this acts like a tiny invisible stray capacitor that lets a small amount of noise leak into the wire.

c) R1 (the main character)

1) What happens if you remove □1? (Simulate the result and explain).

If you remove (R1), the circuit goes haywire whenever the switch is open. Because there is no clean path connecting the wire to the ground, It sucks in random electrical static floating around in the air (the 50Hz wall noise), causing the voltage reading to jump up and down like crazy instead of staying at a flat 0V.

2) What is an input called when a resistor like R1 is missing?

-It is called a floating input (floating node).

3) How should the resistance value for R1 be selected in a real circuit?

-You have to pick a value that balance safety and power saving. If the resistance is too low, it drains your battery way too fast when the switch is closed. If it is too high, it lets outside noise leak in. For real-world gadgets, engineers usually pick a standard (10k ohms) resistor because it is the perfect middle ground.

d) What is the arrow likely pointing to, and what role does □2 serve? What could we call □2?

The Arrow: It simply shows which direction the electrical signal is traveling—it is pointing toward the device or screen that is reading the voltage.

Role of (R2): it represents the resistance of the measuring tool itself (like a voltmeter or computer chip). It isn't a physical resistor you add to the board; it just shows how hard it is for electricity to flow into the brain of the device.

f) What would happen if you brought your hands close to (or touched) an unshielded circuit like this?

If you get close to or touch the wire, the electrical noise on the screen will instantly jump up, because When you touch the unprotected wire, you dump all that static straight into the circuit. It is just like touching an open headphone plug and hearing a loud "**BZZZZ**" sound from the speak

